How Should Urdu Documents Be Chunked for Retrieval? A Budget-Matched Study of Chunk Size, Overlap, Sentence Boundaries and Tokenization

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Abstract

Retrieval systems are shaped by how documents are cut into chunks. Most of what we know about that choice comes from English, and most of it measures size in model tokens. This paper moves the question to Urdu. We build a parallel Urdu–English corpus from UQA and SQuAD2.0: 150 articles and 7,362 test questions. Four multilingual encoders, BM25, and BM25+dense RRF run over five chunkers, word sizes from 50 to 800 (10 to 800 for a bge-m3 extension), token budgets from 64 to 256, and three overlap levels. Budget-matched recall at 500 words is the metric; intervals come from an article-level bootstrap. Three results stand out. The best tested word size is 50 for every encoder in the main grid; extending bge-m3 down to 10 words is monotonic and also selects 50. At 10 words, 18.2% of answers are split across chunks, against 4.1% at 50. The token argmax differs (64/128/256), though adjacent budgets are statistically tied; 128 tokens is about 96 words for XLM-R and 42 for Qwen3 (fertility 1.34 vs. 3.07). Urdu-aware sentence splitting beats English-toolkit defaults by 5.5–6.9 points at 50 words in the four systems where both were run; at 200 words the sign flips. 25% overlap is associated with 2.6–7.2 more points on this corpus in the four systems where it was run, whereas the English studies we reviewed found neutral or negative effects, while orthographic normalization changes nothing measurable on this machine-translated corpus. English is easier on the parallel control but has the same best tested fixed-word size. Code, configurations, and per-question metrics are available at the public repository.

1 Introduction

Urdu has more than 230 million speakers. Retrieval research has not kept up. Mr. TyDi covers eleven languages (Zhang et al., 2021), MIRACL eighteen (Zhang et al., 2022), mMARCO thirteen (Bonifacio et al., 2021); Urdu appears in none of them. The resources that do exist are recent: UQA (Arif et al., 2024), Urdu MS MARCO (Butt et al., 2024), and a small cluster of papers on alignment and Urdu question answering (Faheem et al., 2026; Ahmad et al., 2026). None of them varies how documents are segmented.

Segmentation decides what a retriever can return. English work shows the stakes: size sweeps move recall (Bhat et al., 2025; Bennani and Moslonka, 2026; Wang et al., 2024b), semantic chunking rarely earns its cost (Qu et al., 2024), and structure-aware segmenters help in some settings (Yepes et al., 2024; Singh, 2026). Two questions remain open.

The first is the unit. Nearly every study sizes chunks in model tokens. Tokenizers do not treat languages alike (Petrov et al., 2023; Ahia et al., 2023; Srivastava, 2026). For Urdu the gap is wide: the XLM-R tokenizer behind bge-m3 and multilingual-e5 emits 1.34 tokens per word; Qwen3 emits 3.07. A 256-token chunk is roughly 192 words for one family and 83 for the other. If the optimum is stable in words but not in tokens, specify words. If it is stable in neither, tune per model.

The second question is whether English-toolkit defaults travel. Recursive splitters reach for blank lines, line breaks, spaces, and the ASCII period. Urdu ends sentences with U+06D4 and questions with U+061F; its orthography tolerates Arabic/Persian letter variants, zero-width joiners, and optional diacritics. A language-aware chunker built for Khmer, the only one we found, did worse than plain recursive splitting (Chhoun et al., 2026). Whether that generalises is unknown.

We ask four questions. (Q1) How do size, overlap, and strategy interact with the retriever family (BM25, dense, hybrid)? (Q2) Is the best size stable in words rather than model tokens across tokenizers? (Q3) Do Urdu-aware boundaries and

normalization beat English-toolkit defaults? (Q4) Does the parallel English control select the same best tested size? The evaluation is retrieval-only: no generator, no LLM judge. Our contributions are:

- a systematic, budget-matched chunking evaluation for Urdu retrieval: 360 main runs over a 150-article pool and 7,362 test questions, five chunkers, four encoders, two sparse retrievers, and RRF, plus a four-run bge-m3 lower-size check;
- a direct word-versus-token comparison across tokenizers with measured fertility: token budgets are not comparable across models, word budgets are;
- evidence that Urdu-aware sentence splitting helps at the small sizes dense retrievers prefer and hurts at large sizes; 25% overlap helps on this corpus;
- a parallel English control on the same articles and questions, plus a null result for orthographic normalization on machine-translated Urdu.

2 Related Work

Chunking for retrieval. Fixed token windows are still the default. Bhat et al. (2025) sweep 64–1024 tokens over six English QA datasets; the optimum depends on answer type and encoder. Bennani and Moslonka (2026) vary 50–500 tokens and 0/20% overlap with SPLADE and a generator, and find overlap unhelpful. Wang et al. (2024b) compare 128–2048 tokens inside a larger RAG pipeline. Semantic and structure-aware chunkers have been compared at fixed sizes (Qu et al., 2024; Zhou et al., 2026; Caspari et al., 2026; Merola and Singh, 2025; Günther et al., 2024), and granularity itself has been studied as propositions versus passages (Chen et al., 2023). The closest template is Singh (2026): a factorial study of parsers, chunkers, and embedders on English regulatory documents, analysed with mixed-effects models. We found no controlled chunking evaluation for Urdu retrieval and no study that compares word- and token-defined sizes across tokenizers.

Low-resource and multilingual retrieval. LaBSE (Feng et al., 2020), multilingual E5 (Wang et al., 2024a), BGE-M3 (Chen et al., 2024), and Qwen3 Embedding (Zhang et al., 2025) all claim Urdu support; MMTEB (Enevoldsen et al., 2025) covers a lot of ground. Averages mislead. On Amharic, the best zero-shot multilingual retriever trails a monolingual model by 23% relative MRR, and BM25 beats four of five zero-shot dense retrievers (Alemneh et al., 2026). Hubness and cross-lingual asymmetry distort rankings in other languages (Sakhawat et al., 2026; Goworek et al., 2025). Urdu work has gone into alignment (Faheem et al., 2026), dual-representation retrieval (Ahmad et al., 2026), and QA data (Arif et al., 2024; Butt et al., 2024). We found no controlled study of Urdu segmentation for retrieval.

Tokenization and Urdu text. Tokenizers charge different prices for the same content (Petrov et al., 2023; Ahia et al., 2023), and fertility predicts downstream accuracy in some settings (Lundin et al., 2025). Indic scripts pay a large tax, partly structural (Srivastava, 2026; Regmi et al., 2026); single-token retention exposes how the vocabulary is allocated (Nayeem et al., 2025). Urdu's literature is thinner. CURE built an IR test collection with stemming and query expansion (Iqbal et al., 2020); early indexing work described normalization rules but did not test them (Qureshi et al., 2021). UQA provides machine-translated SQuAD2.0 with preserved spans (Arif et al., 2024); Urdu MS MARCO provides translated passages (Butt et al., 2024). We use UQA because it gives character-level answer spans in parallel with English SQuAD2.0.

3 Data

Parallel corpus. We start from UQA (Arif et al., 2024), a machine translation of SQuAD2.0 into Urdu with preserved answer spans, and from the original SQuAD2.0 articles. The released UQA has 124,745 training rows mapped to 19,021 English paragraphs. One property mattered more than the rest, and we measured it rather than assuming it: each English paragraph was translated again and again. The average paragraph carries 4.86 distinct Urdu contexts; the maximum is 18. Pairs we inspected sit at 0.96–0.98 string similarity.

Documents are built with a *modal-variant* rule. For every English paragraph we keep the most frequent Urdu context (ties go to the longer string, then the lexicographically larger one). A question survives only when its row uses that variant

Articles (dev/test)	150 (30 / 120)
Urdu words	1,007,122
Paragraphs	6,720
Questions (dev/test)	9,077 (1,715 / 7,362)
UQA variants/paragraph	4.86
Rows retained	20.0%
Span failures	0

Table 1: Corpus statistics. Counts are exact, not estimated.

and the question is answerable. An article document is the chosen paragraphs joined in order. Of 124,745 rows, 24,898 remain (20.0%). Every retained span was checked: 83,018/83,018 train and 11,169/11,169 validation rows pass the character-offset equality test. The English control reuses the same articles and questions with SQuAD paragraphs and offsets. On a single-rater 100-question audit, 74 questions were judged self-contained, 21 context-dependent, and 5 unclear; the 95% Wilson interval for the self-contained proportion is 64.6–81.6%. The audit is a corpus check, not an evaluation of retrieval quality.

Pool and splits. We keep the 403 articles with at least 20 retained questions, 20 paragraphs, and 1,500 Urdu words. They are sorted by SHA-1 of the title, and the first 150 are taken. Every fifth article is development (30 articles); the other 120 form the test set. A question-level split would leak chunks across the boundary, so the split is by article. Table 1 gives the totals.

Urdu boundaries and tokenization. Figure 1 shows the linguistic contrast that drives the study: a two-sentence Urdu passage whose word count is the same for every tokenizer but whose token count differs by 3×. The study corpus holds 34,737 U+06D4 boundaries and only 51 U+061F question marks, against 2,310 ASCII periods. Zero-width joiners are absent, tatweel appears once, and there are nine Arabic yeh characters and 34 Urdu digits. Normalization has almost nothing to change (Section 5.4).

4 Method

Chunkers. All five chunkers return character offsets in the normalized document. *Fixed words* slides a window of s whitespace words with step $\max(1, \mathrm{round}(s(1-o)))$. *Fixed tokens* packs whole words while the model tokenizer's content-token count stays within b; it steps by the same overlap fraction in words and never cuts inside a word. *Sentence-aware* splits at U+06D4, U+061F, and the exclamation mark when whitespace or a paragraph break follows, then groups whole sentences up to a word target. Only a sentence longer than twice the target is split by words. *Recursive-English* follows the English-toolkit separators (paragraph, newline, ASCII period followed by whitespace and an uppercase letter or end, question mark, exclamation mark, space) and merges units up to the target. *Recursive-Urdu* runs the same merge with Urdu terminators and paragraph breaks. At zero overlap every chunker covers every non-whitespace character exactly once. We test that on ten documents for every configuration: 1,170 checks, zero uncovered characters.

Retrievers. BM25 tokenizes on whitespace, strips Urdu and ASCII punctuation, and uses $k_1 = 1.2$ and $b = 0.75$. Both values were fixed in advance. BM25-char works on 3–5 character n-grams generated inside words. Dense retrieval is cosine similarity over L2-normalized vectors from bge-m3 (Chen et al., 2024), multilingual-e5-large-instruct (Wang et al., 2024a), Qwen3-Embedding-0.6B (Zhang et al., 2025), and LaBSE (Feng et al., 2020). Queries use the documented instruction formats; the same format is applied to every run. RRF fuses BM25-whitespace and the dense ranking of the same chunk configuration with $k = 60$.

Metrics. A chunk counts as relevant when it contains the full gold answer span. We report Recall@1/3/5/10, MRR@10, nDCG@10, and the primary budget-matched recall. That last one walks the ranking, adds chunk word counts until a budget B is reached, and scores 1 if at least one retrieved chunk contains the full answer. A span split across chunks is a failure, and we report the answer-split rate next to recall. The primary budget is $B = 500$ words; 200, 1000, and 2000 are secondary. The crossing definition, which includes the chunk that reaches the budget, is the pre-registered one. The strict definition appears in the appendix; it changes only the BM25 argmax. Budget-matched here means the same 500-word query-time reading budget for every configuration. Index size, chunk count, and embedding cost are not equalised; they are reported so the trade-off is visible.

Statistics. The analysis unit is the article, not the question. Every question in an article shares the same chunk set, so averaging over questions

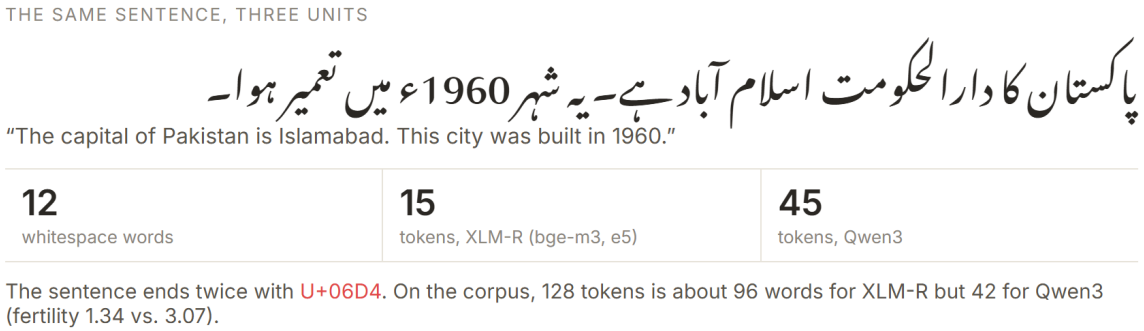


Figure 1: The same 12-word Urdu sentence is 15 tokens for the XLM-R tokenizer used by bge-m3 and multilingual-e5, and 45 tokens for Qwen3. Sentence ends are U+06D4. On the corpus, 128 tokens corresponds to about 96 words for XLM-R and 42 for Qwen3.

would pretend to a precision the design does not have. For each configuration we compute a per-article mean, then average over the 120 test articles. Confidence intervals come from 10,000 bootstrap resamples over articles with seed 0. Paired contrasts reuse the same resample for both configurations. The arithmetic is blunt: 61 questions share one chunk set, and treating them as independent reports power 1.00 for a 5-point effect that the article-level analysis detects with about 75% power.

Grid. BM25 covers five strategies, five word sizes (50/100/200/400/800), and three overlaps (0/10/25%). Dense models cover 21 configurations: fixed words at 50/100/200 with 0/10/25% overlap (9), fixed tokens at 64/128/256 (3), and sentence-aware, recursive-English, and recursive-Urdu at 50/100/200 with zero overlap (9). bge-m3 and Qwen3 also run 400/800 words with 0/25% overlap. LaBSE is secondary and runs at 50/100 words. e5 stops at 200 words: its 512-token window truncates 74% of 400-word chunks and 100% of 800-word chunks. The encoders use their model maximum sequence lengths (bge-m3 8,192, e5 512, Qwen3 32,768, LaBSE 256). Only LaBSE's 100-word sentence-aware run truncates: 3 of 9,800 chunks (0.03%) exceed 256 tokens; every other dense run is clean. The main grid is 360 runs. A separate lower-size extension adds 4 bge-m3 runs at 10/25/40/50 words (overlap 0, N0), reported in §5.2.

5 Results

5.1 Size and retriever

Dense and sparse retrievers disagree about size (Figure 2, Table 2). Every dense encoder does best at 50 words in the tested range and declines thereafter, with no second peak. BM25 climbs to 200–400 words and drops at 800. At 50 words bge-m3 reaches 0.773 (95% CI 0.758–0.789), 6.5 points above BM25's best tested 0.708 (0.688–0.729). BM25's best tested size is four to eight times longer than the dense size.

Model	50	100	200	400	800
bge-m3	**.773**	.757	.680	.570	.415
e5-large	**.776**	.752	.663	—	—
Qwen3-0.6B	**.603**	.556	.476	.388	.272
LaBSE	**.540**	.487	—	—	—
BM25	.648	.684	**.708**	**.709**	.612

Table 2: BM-Recall@500 by word size (overlap 0, bm500x). Bold is the row maximum; 200 and 400 are statistically tied for BM25. Full CIs are in Figure 2.

The fall is steep. bge-m3 loses 1.6 points from 50 to 100 and 7.7 from 100 to 200; e5 loses 2.4 and 8.9; Qwen3 loses 4.7 and 8.0. Every encoder's 50–100 drop is significant (bge +1.65 [0.53, 2.80], e5 +2.40 [1.06, 3.71], Qwen3 +4.71 [3.35, 6.11], LaBSE +5.38 [3.89, 6.86]).

5.2 Lower-size check

The main grid stops at 50 words for dense models. To test whether 50 is an interior maximum, we extended bge-m3 down to 10 words on the same articles and questions (zero overlap, N0). Table 3 reports the four runs. BM-Recall@500 rises monotonically: .418 at 10, .689 at 25, .756 at 40, and .773 at 50. The 50-word point is 1.7 points above 40, but the two intervals overlap, so we do not claim a sharp 40-versus-50 separation. At 10 words, 18.2% of answers are split across chunks, against 4.1% at 50; fragmentation is a plausible mechanism, not a decomposition. The check cov-

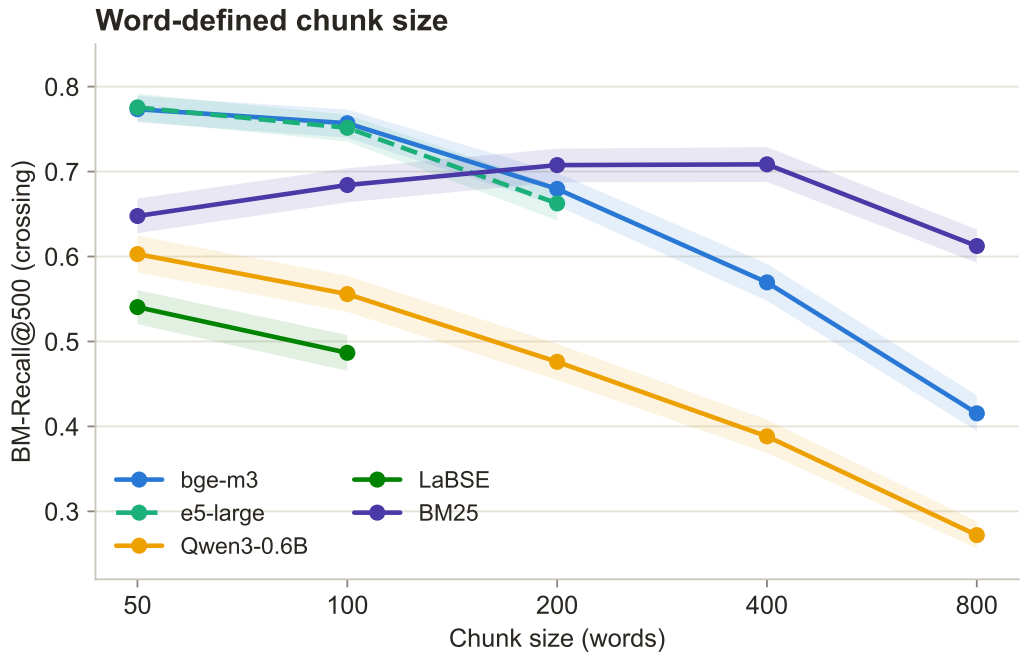


Figure 2: BM-Recall@500 (crossing budget) against chunk size in words. Bands are 95% article-level bootstrap CIs. Dense encoders peak at the smallest tested size; BM25 peaks at 200–400 words. bge-m3 and e5 are nearly coincident.

Size	Chunks	BM500x [95% CI]	BM500	Split
10	80,962	.418 [.400, .437]	.417	18.2%
25	32,423	.689 [.672, .707]	.686	7.7%
40	20,285	.756 [.741, .771]	.753	4.8%
50	16,244	**.773** [.758, .789]	.768	4.1%

Table 3: bge-m3 lower-size check (10/25/40/50 words, zero overlap, N0, crossing budget). BM500x is the primary metric; Split is the answer-split rate. This is one encoder, not the full grid.

Model	64	128	256	words at argmax
bge-m3	.767	**.768**	.677	96
e5-large	**.765**	.762	.658	48
Qwen3-0.6B	.489	.589	**.590**	83

Table 4: BM-Recall@500 by token budget (top row: tokens; bottom row: implied whitespace words at the selected budget).

ers one encoder. e5, Qwen3, and LaBSE were not extended below 50.

5.3 Words versus model tokens

All four encoders' best tested word size is 50. Their token argmaxes do not agree: 128 for bge-m3, 64 for e5, 256 for Qwen3 (Table 4). The disagreement is softer than it looks. Neighbours are statistically tied: bge 64 vs. 128 is -0.08 points $[-1.60, 1.43]$; e5 is $+0.35$ $[-1.08, 1.75]$; Qwen3 128 vs. 256 is -0.09 $[-1.54, 1.33]$. These are plateaus, not sharp breaks.

What a token means is the durable difference. At 128 tokens XLM-R sees about 96 words; Qwen3 sees about 42 (Figure 3). At 256 tokens the gap is roughly 192 words against 83. Word sizes remove that ambiguity. Token sizes quietly change the task from one model to the next.

5.4 Boundaries and normalization

At 50 words, sentence-aware splitting beats the English recursive default in all four systems where both were run (LaBSE was not run with recursive-English; Figure 4): bge-m3 +6.02 [4.98, 7.06], e5 +6.21 [5.18, 7.21], Qwen3 +6.90 [5.59, 8.15], BM25 +5.53 [4.31, 6.70]. Against fixed-size chunks the gain is larger: +11.4, +11.0, +12.7, and +10.1 points. At 100 words it shrinks and turns mixed (bge +1.41 [0.30, 2.53]; e5 +0.86 [−0.32, 1.99]; Qwen3 −2.04 [−3.34, −0.74]). At 200 it reverses: bge −8.99, e5 −10.38, Qwen3 −11.75, BM25 −3.09.

The answer-split rate shows part of the mechanism. Fixed 50-word chunks cut 4.10% of answers; sentence-aware 50-word chunks cut 0.05%. At 200 words the rates are 1.06% and 0.00%. The reversal is therefore not about splitting. The English recursive rule, applied to Urdu, produces

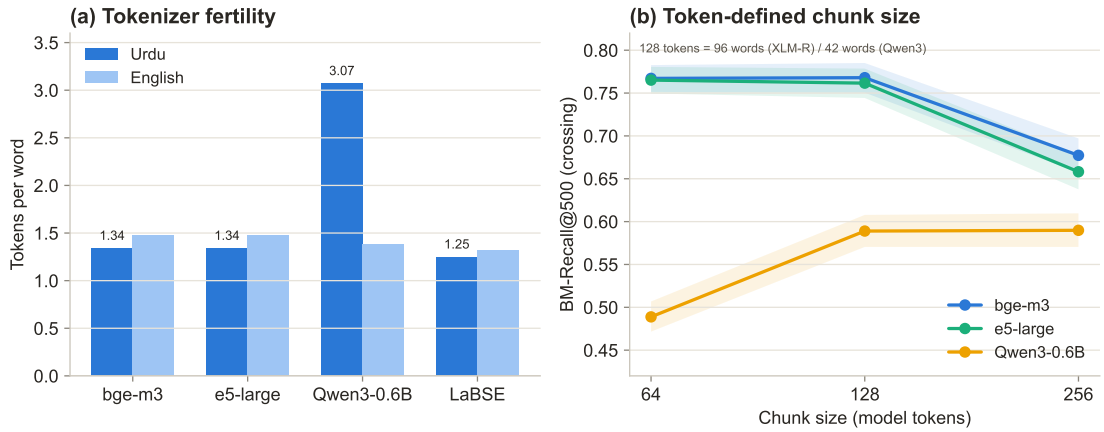


Figure 3: (a) Tokens per whitespace word by tokenizer. (b) BM-Recall@500 against token budget. Bands are 95% article-level bootstrap CIs. Adjacent token budgets are statistically tied; the same budget is a different amount of text for each tokenizer.

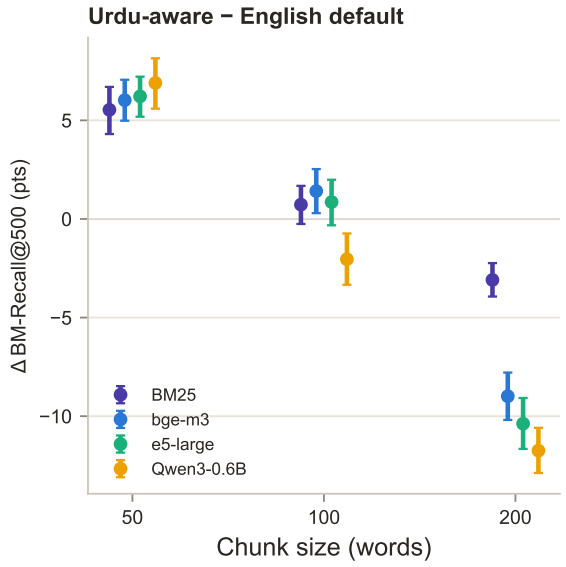


Figure 4: Difference between Urdu-aware sentence splitting and the English recursive default (bm500x, points). Positive means Urdu-aware is better. The advantage at 50 words reverses at 200.

smaller units and a different retrieval granularity at large targets. This is an interaction, not a free win.

Normalization barely moves. N1 and N2 differ from N0 by at most 0.09 points. Every article-level CI covers zero except a 0.085-point N2 gain for BM25 at 50 words ($p = 0.038$, uncorrected and exploratory). The Unicode audit explains why: no ZWNJ or ZWJ, one tatweel, nine Arabic yeh characters, 34 Urdu digits. We report the null instead of dressing it up.

5.5 Overlap

Against the English studies we reviewed, 25% overlap helps at every tested size in the four systems where overlap was run (LaBSE was not; Figure 5). bge-m3 gains 6.91 [5.72, 8.05] at 50 words, 5.24 [4.29, 6.18] at 100, and 4.22 [3.10, 5.38] at 200. e5 gains 7.16, 5.13, and 3.77; Qwen3 6.13, 5.75, and 4.97. BM25 follows with 6.97, 5.51, and 2.60. Overlap keeps spans intact and gives 50-word chunks enough surrounding text to rank. Both effects are visible in the per-question data. The result is corpus-specific; we do not claim it travels.

5.6 English control

English is easier on the parallel questions, but the best tested fixed-word size does not move: 50 words in both languages (Table 5). The gap is 7.4 points at 50, 7.2 at 100, and 7.8 at 200. For recursive-English chunks it is 10.8 at 50, 6.3 at 100, and gone at 200 (-0.9 [-2.29, 0.46]). Nothing here says the languages need different sizes. It does say the Urdu side is harder, which fits translation artifacts and weaker Urdu coverage in the encoders.

5.7 Best systems

Sentence-aware 50-word chunks with bge-m3 score 0.888 [0.876, 0.899]. e5 is 0.886 [0.873, 0.897]; the two are tied. RRF reaches 0.871 [0.857, 0.884] with bge and 0.865 with e5, below the best single model. It does lift the weak ones: Qwen3 from 0.730 to 0.814, LaBSE from 0.703 to 0.810, BM25 from 0.749 to 0.871. BM25-

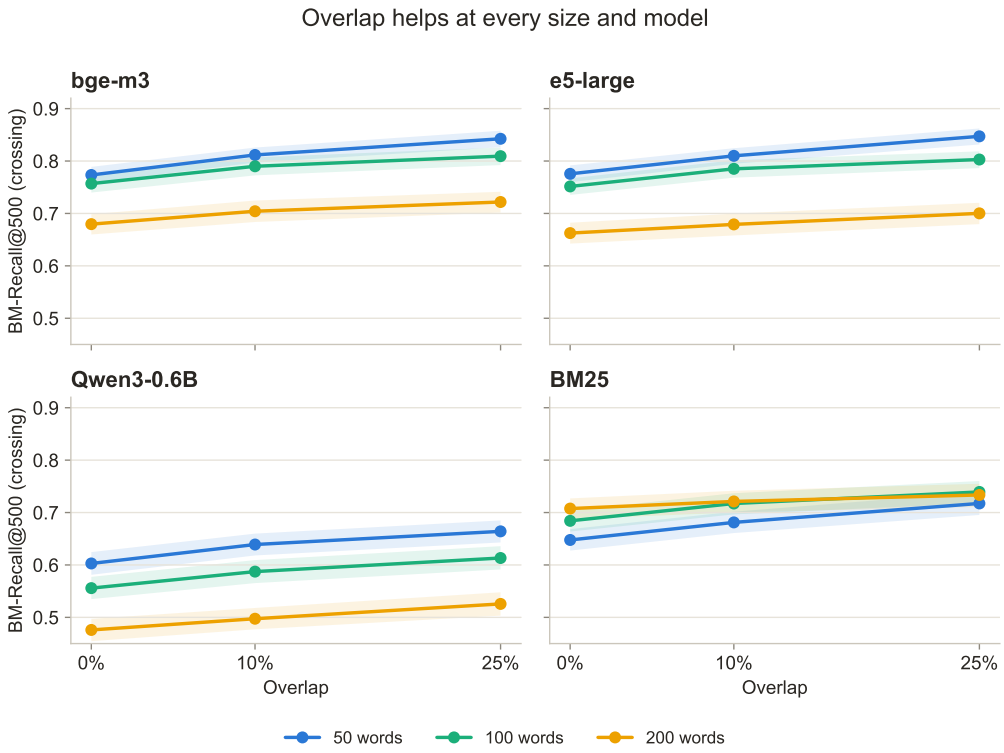


Figure 5: BM-Recall@500 against overlap for four systems and three sizes. Bands are 95% article-level bootstrap CIs. Overlap helps everywhere; the effect is largest at 50 words.

Strategy	Size	Urdu	English	Diff.
fixed words	50	.773	.847	+7.4
fixed words	100	.757	.829	+7.2
fixed words	200	.680	.758	+7.8
recursive English	50	.828	.935	+10.8
recursive English	100	.836	.899	+6.3
recursive English	200	.809	.800	−0.9

Table 5: Parallel English control with bge-m3. Diff. is English minus Urdu in points; the 200-word recursive difference is not significant.

System	BM-Recall@500
bge-m3, sentence-aware 50	**.888** [.876, .899]
e5-large, sentence-aware 50	.886 [.873, .897]
bge-m3 + RRF	.871 [.857, .884]
BM25, sentence-aware 50	.749 [.728, .769]
Qwen3-0.6B, sentence-aware 50	.730 [.710, .750]
LaBSE, sentence-aware 50	.703 [.683, .723]

Table 6: Best configuration per system (95% article-level bootstrap CIs).

char trails BM25-whitespace by 2–4 points at every size. Table 6 has the full ranking.

6 Discussion

Words are the portable unit; tokens are not. Specify chunk size in words when encoders use different tokenizers. A word is a language unit. A token is a model unit whose size depends on the vocabulary. The measured fertility gap here is 2.3× (1.34 vs. 3.07). That is enough to turn one nominal configuration into two different retrieval tasks. The token argmaxes are plateaus, so the problem is not a sharp 64-versus-256 disagreement. It is that "128 tokens" does not name a fixed amount of text.

Boundary for three encoders; bge-m3 confirmed downward. Above 50 words every dense curve falls. For e5, Qwen3, and LaBSE, 50 is the smallest size tested. For bge-m3, 10–50 are now tested and 50 is still best; the curve rises monotonically into 50. The 40- and 50-word intervals overlap, so the lower end is not sharply separated, and 30/35/45 remain untested. The data show that the best *tested* size is the same across encoders; they do not show that 50 is optimal in any absolute sense. The boundary also explains why Urdu-aware splitting gains most at 50 words: sentence-aware chunks avoid the 4.1% answer-split rate of fixed 50-word chunks.

Why overlap helps here. English chunking work often finds overlap neutral or harmful (Ben-

nani and Moslonka, 2026; Amiri and Bocklitz, 2025). This corpus differs in two ways. Documents are concatenations of short translated paragraphs, and answers are short spans. Overlap keeps a span intact when a boundary would cut it, and it gives 50-word chunks context they otherwise lack. Four systems show the same direction, so it is not one model's quirk. It remains a corpus-specific result.

Language-aware splitting is conditional. Urdu-aware splitting is not a free win. At 50 words it beats the English default by six points. At 200 words it loses 3 points for BM25 and 9–12 for the dense encoders. A paper that reported one size could reach the opposite conclusion. The Khmer result fits this pattern (Chhoun et al., 2026): language-aware chunkers interact with size and with the recursive merge policy. One size is not a test.

Normalization is a property of the corpus. The null is informative. UQA's translation pipeline already produces normalized Urdu: no ZWNJ, essentially no tatweel, almost all digits ASCII (34 are Urdu digits), and nine Arabic-variant yeh characters. A study on naturally occurring Urdu web text could still find an effect. Here there is almost nothing to normalize. We report the Unicode audit so the null can be read rather than guessed at.

7 Limitations

- **Machine-translated corpus.** UQA is MT, with 4.86 variants per paragraph. The modal rule keeps 20% of rows and may select a systematically different translation. The English control is original SQuAD text, so the language comparison mixes language with translation artifacts.
- **Boundary result.** 50 words is the smallest size tested for e5, Qwen3, and LaBSE. bge-m3 was extended to 10/25/40/50; 50 remains best and the curve is monotonic, but 30/35/45 are untested and no interior maximum is identified.
- **Single domain.** Wikipedia articles made of concatenated paragraphs. Not web text, legal text, or conversation.
- **Machine-translated queries.** Every question is UQA machine translation. The 150–300 native-authored questions needed to test transfer do not exist. A single-rater audit judges 74 of 100 sampled questions self-contained, 21 context-dependent, and 5 unclear (Wilson 95% CI for the self-contained rate 64.6–81.6%); no second-reader agreement estimate is available.
- **Retrieval only.** No generator and no LLM judge. The conclusions concern evidence retrieval, not answer quality. Relevance is exact answer-string containment; no human relevance or context-quality judgments were collected.
- **Selection and multiplicity.** Confidence intervals are marginal, not adjusted for the many configurations, and the best system in Table 6 was selected after inspecting all runs; its interval is optimistically narrow. The one $p = 0.038$ normalization contrast would not survive a strict multiplicity correction.
- **Fixed BM25.** k_1 and b were fixed in advance, not tuned. A tuned BM25 is a different experiment.
- **Small normalization exclusions.** 37 of 9,077 questions (0.41%) fail N1/N2 span mapping because UQA answers are whitespace-padded. They are excluded from those variants only.
- **Prefix sensitivity.** Qwen3's documented instruction prefix lowered BM500 relative to no prefix (0.400 vs. 0.460). Its absolute numbers may understate the model.

8 Ethics and Reproducibility

SQuAD2.0 is CC BY-SA 4.0. UQA has no explicit licence, so the public artifact contains code, configurations, question IDs, and per-question metrics rather than UQA text. It is available at the public repository:

github.com/devxyasir/urdu-chunking-strategy

The corpus is encyclopedic and contains no personal or sensitive data. No LLM generated evaluation data, labels, or rankings. Every question and span comes from the released corpora, and every metric is deterministic. We report 360 main runs and 4 lower-size runs, bootstrap seed 0, model revisions, per-question metrics, and article-level CIs. AI assistance was used for literature search,

code development, analysis, figure generation, and drafting/editing. The author verified all results and takes responsibility for the content.

9 Conclusion

For Urdu retrieval, define chunk size in words, not model tokens. The same token budget is 96 words for XLM-R and 42 for Qwen3, while the best tested word size is 50 for every encoder; extending bge-m3 down to 10 words does not move it. Urdu-aware sentence splitting helps at the sizes dense retrievers prefer (5.5–6.9 points) and reverses at 200 words. 25% overlap helps on this corpus. Normalization is a null on machine-translated Urdu. We release the pipeline and per-question results so the size-by-unit interaction can be tested on other languages and on naturally occurring text.

References

Orevaoghene Ahia, Sachin Kumar, Hila Gonen, Jungo Kasai, David R. Mortensen, Noah A. Smith, and Yulia Tsvetkov. 2023. Do all languages cost the same? tokenization in the era of commercial language models. *Preprint*, arXiv:2305.13707v1.

Saad Ahmad, Muhammad Hammad, Muhammad Zeeshan, Faizad Ullah, and Asim Karim. 2026. DR-RAG: Addressing retrieval misalignment in low-resource Urdu question answering. In *Proceedings of the Second Workshop on Challenges in Processing South Asian Languages (CHiPSAL 2026)*, pages 49–58, Palma, Mallorca, Spain. ELRA Language Resources Association (ELRA).

Yosef Worku Alemneh, Kidist Amde Mekonnen, and Maarten de Rijke. 2026. The multilingual curse at the retrieval layer: Evidence from amharic. *Preprint*, arXiv:2605.24556v1.

Mahmoud Amiri and Thomas Bocklitz. 2025. Chunk twice, embed once: A systematic study of segmentation and representation trade-offs in chemistry-aware retrieval-augmented generation. *Preprint*, arXiv:2506.17277v2.

Samee Arif, Sualeha Farid, Awais Athar, and Agha Ali Raza. 2024. Uqa: Corpus for urdu question answering. *Preprint*, arXiv:2405.01458v2.

Sofia Bennani and Charles Moslonka. 2026. A systematic analysis of chunking strategies for reliable question answering. *Preprint*, arXiv:2601.14123v1.

Sinchana Ramakanth Bhat, Max Rudat, Jannis Spiekermann, and Nicolas Flores-Herr. 2025. Rethinking chunk size for long-document retrieval: A multi-dataset analysis. *Preprint*, arXiv:2505.21700v2.

Luiz Bonifacio, Vitor Jeronymo, Hugo Queiroz Abonizio, Israel Campiotti, Marzieh Fadaee, Roberto Lotufo, and Rodrigo Nogueira. 2021. mmarco: A multilingual version of the ms marco passage ranking dataset. *Preprint*, arXiv:2108.13897v5.

Umer Butt, Stalin Varanasi, and Günter Neumann. 2024. Enabling low-resource language retrieval: Establishing baselines for urdu ms marco. *Preprint*, arXiv:2412.12997v3.

Laura Caspari, Kanishka Ghosh Dastidar, Michael Dinzinger, Jelena Mitrović, and Michael Granitzer. 2026. When is complex chunking worth it? a multi-objective evaluation of chunking methods at scale. *Preprint*, arXiv:2608.16586v1.

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *Preprint*, arXiv:2402.03216v5.

Tong Chen, Hongwei Wang, Sihao Chen, Wenhao Yu, Kaixin Ma, Xinran Zhao, Hongming Zhang, and Dong Yu. 2023. Dense x retrieval: What retrieval granularity should we use? *Preprint*, arXiv:2312.06648v3.

Sovandara Chhoun, Pichdara Po, Sereiwathna Ros, Wan-Sup Cho, and Saksonita Khoeurn. 2026. Evaluation of chunking strategies for effective text embedding in low-resource language on agricultural documents. *Preprint*, arXiv:2605.22203v1.

Kenneth Enevoldsen, Isaac Chung, Imene Kerboua, Márton Kardos, Ashwin Mathur, David Stap, Jay Gala, Wissam Siblini, Dominik Krzemiński, Genta Indra Winata, Saba Sturua, Saiteja Utpala, Mathieu Ciancone, Marion Schaeffer, Gabriel Sequeira, Diganta Misra, Shreeya Dhakal, Jonathan Rystrøm, Roman Solomatin, and 67 others. 2025. Mmteb: Massive multilingual text embedding benchmark. *Preprint*, arXiv:2502.13595v4.

Ali Faheem, Muhammad Hammad, Faizad Ullah, Ahmed Hassan, Fezan Rasool, and Asim Karim. 2026. Beyond fine-tuning: Procrustes alignment of multilingual embeddings for low-resource cross-lingual retrieval. In *Proceedings of the SIGUL 2026 Joint Workshop with ELE, EURALI, and DCLRL: Towards Inclusivity and Equality: Language Resources and Technologies for Under-Resourced and Endangered Languages*, pages 233–241, Palma, Mallorca, Spain. ELRA Language Resources Association (ELRA).

Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2020. Language-agnostic bert sentence embedding. *Preprint*, arXiv:2007.01852v2.

Roksana Goworek, Olivia Macmillan-Scott, and Eda B. Özyiğit. 2025. What drives cross-lingual ranking? retrieval approaches with multilingual language models. *Preprint*, arXiv:2511.19324v1.

Michael Günther, Isabelle Mohr, Daniel James Williams, Bo Wang, and Han Xiao. 2024. Late chunking: Contextual chunk embeddings using long-context embedding models. *Preprint*, arXiv:2409.04701v3.

Muntaha Iqbal, Kamran Amjad, Bilal Tahir, and Muhammad Amir Mehmood. 2020. Cure: Collection for urdu information retrieval evaluation and ranking. *Preprint*, arXiv:2011.00565v1.

Jessica M. Lundin, Ada Zhang, Nihal Karim, Hamza Louzan, Victor Wei, David Adelani, and Cody Carroll. 2025. The token tax: Systematic bias in multilingual tokenization. *Preprint*, arXiv:2509.05486v1.

Carlo Merola and Jaspinder Singh. 2025. Reconstructing context: Evaluating advanced chunking strategies for retrieval-augmented generation. *Preprint*, arXiv:2504.19754v1.

Mir Tafseer Nayeem, Sawsan Alqahtani, Md Tahmid Rahman Laskar, Tasnim Mohiuddin, and M Saiful Bari. 2025. Beyond fertility: Analyzing strr as a metric for multilingual tokenization evaluation. *Preprint*, arXiv:2510.09947v2.

Aleksandar Petrov, Emanuele La Malfa, Philip H. S. Torr, and Adel Bibi. 2023. Language model tokenizers introduce unfairness between languages. *Preprint*, arXiv:2305.15425v2.

Renyi Qu, Ruixuan Tu, and Forrest Bao. 2024. Is semantic chunking worth the computational cost? *Preprint*, arXiv:2410.13070v1.

Muhammad Mudassar Qureshi, Muhammad Shoaib, and Kalsoom. 2021. An efficient indexing and searching technique for information retrieval for urdu language. *Preprint*, arXiv:2103.00532v1.

Sajal Regmi, Siddhartha Pudasaini, and Chetan Phakami Pun. 2026. Vowel signs are not letters: A pre-tokenization ceiling on multilingual tokenizer fertility. *Preprint*, arXiv:2608.26449v1.

Adib Sakhawat, Fardeen Sadab, and Atik Shahriar. 2026. Hubness, not anisotropy, drives cross-lingual retrieval asymmetry in multilingual embedding models. *Preprint*, arXiv:2605.26575v1.

Shubham Kumar Singh. 2026. Parser, chunking, and embedding interactions in retrieval-augmented generation over indian government regulatory documents. *Preprint*, arXiv:2609.31660v1.

Priyansh Srivastava. 2026. The tokenizer tax: Quantifying and explaining the cross-lingual cost of subword tokenization for indian languages. *Preprint*, arXiv:2607.24276v1.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024a. Multilingual e5 text embeddings: A technical report. *Preprint*, arXiv:2402.05672v1.

Xiaohua Wang, Zhenghua Wang, Xuan Gao, Feiran Zhang, Yixin Wu, Zhibo Xu, Tianyuan Shi, Zhengyuan Wang, Shizheng Li, Qi Qian, Ruicheng Yin, Changze Lv, Xiaoqing Zheng, and Xuanjing Huang. 2024b. Searching for best practices in retrieval-augmented generation. *Preprint*, arXiv:2407.01219v1.

Antonio Jimeno Yepes, Yao You, Jan Milczek, Sebastian Laverde, and Renyu Li. 2024. Financial report chunking for effective retrieval augmented generation. *Preprint*, arXiv:2402.05131v3.

Xinyu Zhang, Xueguang Ma, Peng Shi, and Jimmy Lin. 2021. Mr. tydi: A multi-lingual benchmark for dense retrieval. *Preprint*, arXiv:2108.08787v2.

Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. 2022. Making a miracl: Multilingual information retrieval across a continuum of languages. *Preprint*, arXiv:2210.09984v1.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *Preprint*, arXiv:2506.05176v3.

Yongjie Zhou, Shuai Wang, Bevan Koopman, and Guido Zuccon. 2026. Beyond chunk-then-embed: A comprehensive taxonomy and evaluation of document chunking strategies for information retrieval. *Preprint*, arXiv:2602.16974v1.

A Additional results

Strict versus crossing budget. The main text uses the crossing definition. Under the strict definition the BM25 argmax moves from 200 to 100 words (0.686 at 100, 0.663 at 200). Every dense argmax stays at 50. No dense conclusion changes.

Normalization. N0/N1/N2 means for BM25-whitespace at 50/100/200 words differ by at most 0.09 points. The largest difference is N2 at 50 words (+0.085, $p = 0.038$, uncorrected and exploratory). N1/N2 exclude 37 whitespace-padded answers.

Prefix checks. On 100 dev questions at 200 words, the e5 instruction prefix improves BM500 from 0.550 to 0.640 and R@1 from 0.43 to 0.49. The Qwen3 prefix changes BM500 from 0.460 to 0.400 and R@1 from 0.31 to 0.32. Both models use their documented format in every reported run.

Grid. BM25: 2 sparse variants × (75 base configurations + 6 N1/N2 normalization runs) = 162 runs. Dense: 54 for bge-m3 (21 Urdu + 12 at 400/800 + 21 English), 21 for e5, 33 for Qwen3 (21 + 12 at 400/800), and 6 for LaBSE = 114 runs. RRF: 84 runs. Total 162 + 114 + 84 = 360. The 21-configuration base is fixed words at 50/100/200 with 0/10/25% overlap (9), fixed tokens at 64/128/256 (3), and sentence-aware, recursive-English, and recursive-Urdu at 50/100/200 with zero overlap (9). The lower-size check adds 4 bge-m3 runs at 10/25/40/50 words (zero overlap, N0), for 364 runs including the extension. The check is reported separately in §5.2; it is not part of the archived 360-run package.

Model revisions and environment.

Model revisions: bge-m3
5617a9f61b028005a4858fdac845db406aefb181
e5 274baa43b0e13e37fafa6428dbc7938e62e5c439
Qwen3 97b0c614be4d77ee51c0cef4e5f07c00f9eb65b3
LaBSE
836121a0533e5664b21c7aacc5d22951f2b8b25b

The archived runs used sentence-transformers 6.2.0, transformers 5.20.2, torch 2.11.0+cu130, CUDA 13.0, and one NVIDIA A100-SXM4-40GB. The bootstrap uses seed 0. The manifest's budget_mode_primary key records a hard-coded label from an earlier draft; the primary metric in every table is bm500x (crossing), and bm500 is the strict variant.